Notes: Wermers (JF 2000)

Mutual Fund Performance: An Empirical Decomposition into

Stock-Picking Talent, Style, Transactions Costs, and Expenses

February 1, 2026

Contents

1 Executive takeaways and research question

The central question is whether *active mutual fund management adds value.* Instead of relying only on return regressions, the paper merges a holdings database (CDA) with mutual-fund return/characteristics data (CRSP) to compute:

- **What funds actually hold and how those holdings perform** (a *holdings-based* "gross" stock-portfolio return),
- **What investors actually receive** (the realized *net* mutual fund return),
- and then a **decomposition** that separates skill-related components (stock selection and timing) from mechanical components (style exposures, expenses, and execution costs).

The key punchline is economically sharp: funds *pick stocks that outperform* the market, but investors do *not* receive that outperformance after accounting for nonstock holdings, expenses, and trading costs.

2 Data construction and basic objects

2.1 Merged holdings–returns database (CDA $\times$ CRSP)

The sample covers U.S. equity mutual funds over 1975–1994, created by matching:

- **CDA** quarterly snapshots of equity holdings (stock-level positions and total net assets),
- **CRSP** mutual-fund net returns and characteristics (including expense ratios and turnover).

The analysis focuses on diversified equity mutual funds (with self-declared objectives such as aggressive growth, growth, growth & income, income, and balanced), and it constructs both:

1. **TNA-weighted** (value-weighted by total net assets) averages, which describe the experience of a representative dollar invested in the fund industry, and
2. **equal-weighted** averages, which emphasize the typical fund rather than the typical invested dollar.

2.2 Key return definitions: net vs. holdings-based "gross"

Let a quarter index be t. For each fund, the holdings database provides portfolio weights at the end of quarter $t-1$. A holdings-based (buy-and-hold) **gross stock-portfolio return** for quarter t is computed as

$$R_t^{\text{stock}} \equiv \sum_{j=1}^{N} \tilde{w}_{j,t-1}\, \tilde{R}_{j,t}, \tag{1}$$

where $\tilde{w}_{j,t-1}$ is the weight on stock j at the end of $t-1$ (normalized to sum to one across stocks) and $\tilde{R}_{j,t}$ is the buy-and-hold return of stock j during quarter t.

The CRSP database provides the realized **net mutual fund return** (net of reported fund expenses and inclusive of trading costs through NAV changes) for each fund and month; yearly returns are formed by compounding quarterly rebalanced measures.

2.3 Characteristic-matched benchmark portfolios (the style benchmark)

To separate "picking" from "style", the paper uses the Daniel et al. (1997) characteristic benchmark procedure:

- At each June, stocks are sorted into size quintiles.
- Within each size quintile, stocks are sorted into book-to-market quintiles (using book-to-market measured at the prior December).
- Within each size–value cell, stocks are sorted into momentum quintiles (past 12-month returns through May).

This yields $5 \times 5 \times 5 = 125$ benchmark portfolios. For each stock j, denote by $\tilde{R}^{b}_{t,j,t-1}$ the quarter-t return of the benchmark portfolio matched to stock j using characteristics measured at the end of quarter $t-1$.

3 Performance decomposition models (the core "models" of the paper)

3.1 (1) Characteristic Selectivity: stock-picking skill controlling for style

The **Characteristic Selectivity** measure (CS) is the holdings-based, characteristic-adjusted return of the fund's stock portfolio (paper Eq. (1)):

$$\mathrm{CS}_t \;=\; \sum_{j=1}^{N} \tilde{w}_{j,t-1}\Big(\tilde{R}_{j,t} - \tilde{R}^{b}_{t,j,t-1}\Big). \tag{2}$$

Interpretation. CS_t is positive when the manager holds stocks that outperform their characteristic-matched benchmarks. This aims to isolate *selection* beyond size/value/momentum tilts.

3.2 (2) Characteristic Timing: timing the payoffs to characteristics

The stock-selection measure above does not capture a manager's ability to *time* characteristics (e.g., increasing momentum exposure when momentum is about to pay off). The **Characteristic Timing** measure (CT) uses changes in holdings weights to detect whether the manager tilted toward characteristics that subsequently performed well (paper Eq. (2)):

$$\mathrm{CT}_t \;=\; \sum_{j=1}^{N}\Big(\tilde{w}_{j,t-1}\tilde{R}^b_{t,j,t-1} - \tilde{w}_{j,t-5}\tilde{R}^b_{t,j,t-5}\Big). \tag{3}$$

Here $t-5$ means "one year earlier" in quarterly data (four quarters plus the end-of-quarter indexing convention).

Interpretation. If the manager increases weight on stocks associated with benchmark portfolios that earn high returns in quarter t, then CT_t is large.

3.3 (3) Average Style: the long-run return from characteristic tilts

The **Average Style** component (AS) measures the portion of the stock-portfolio return attributable to the fund's longer-run tendency to hold stocks with particular characteristics (paper Eq. (3)):

$$\mathrm{AS}_t \;=\; \sum_{j=1}^{N}\tilde{w}_{j,t-5}\tilde{R}^b_{t,j,t-5}. \tag{4}$$

Interpretation. AS_t captures the return that would be earned by passively holding the fund's *one-year-lagged* characteristic exposures, thus abstracting from short-horizon timing.

3.4 Identity: why CS + CT + AS reconstructs the gross stock return

A critical internal consistency check is that these three components add up to the holdings-based gross stock return in equation (1). Indeed,

$$\begin{aligned}\mathrm{CS}_t + \mathrm{CT}_t + \mathrm{AS}_t &= \sum_j \tilde{w}_{j,t-1}(\tilde{R}_{j,t} - \tilde{R}^b_{t,j,t-1}) + \sum_j\big(\tilde{w}_{j,t-1}\tilde{R}^b_{t,j,t-1} - \tilde{w}_{j,t-5}\tilde{R}^b_{t,j,t-5}\big) + \sum_j \tilde{w}_{j,t-5}\tilde{R}^b_{t,j,t-5}\\ &= \sum_j \tilde{w}_{j,t-1}\tilde{R}_{j,t} \;=\; R^{\text{stock}}_t.\end{aligned}$$

Thus, **gross stock-portfolio return** can be read as:

$$\text{Gross stock return} = \text{Selection} + \text{Timing} + \text{Style}.$$

3.5 Execution-cost model: estimating transaction costs from holdings changes

Because fund *net* returns reflect trading frictions while the holdings-based gross return does not, the paper estimates trading (execution) costs using empirical models from Keim–Madhavan (1997) and time-series adjustments from Stoll (1995).

For a **buy** trade of stock i during quarter t, the estimated percentage cost is (paper Eq. (4)):

$$C^B_{i,t} = Y^k_t\Big[1.098 + 0.336D^{Nasdaq}_{i,t} + 0.092\,Trsize_{i,t} - 0.084\log(mcap_{i,t}) + 13.807\Big(\frac{1}{P_{i,t}}\Big)\Big]. \tag{5}$$

For a **sell** trade, the estimated percentage cost is (paper Eq. (5)):

$$C^S_{i,t} = Y^k_t \Big[0.979 + 0.058 D^{Nasdaq}_{i,t} + 0.214\, Trsize_{i,t} - 0.059 \log(mcap_{i,t}) + 6.537\Big(\frac{1}{P_{i,t}}\Big)\Big]. \tag{6}$$

Definitions:

- $D^{Nasdaq}_{i,t} = 1$ if the stock trades on Nasdaq; 0 otherwise.
- $Trsize_{i,t}$ is the dollar size of the trade scaled by the stock's market capitalization.
- $mcap_{i,t}$ is the stock's market capitalization (in \$thousands in the original calibration).
- $P_{i,t}$ is the stock price.
- Y^k_t is a *year–market factor* (market $k \in$ {NYSE/AMEX, Nasdaq}) capturing time variation in trading costs.

Fund-level transaction costs are obtained by applying these trade-level cost estimates to inferred trades from changes in holdings, summing across trades in the quarter, and scaling by the fund's stock-portfolio value.

3.6 Regression-based risk-adjusted performance: Carhart (and Jensen) alphas

To compare holdings-based measures to classic performance evaluation, the paper also uses factor-model alphas. The **Carhart four-factor model** is (paper Eq. (6)):

$$R_{j,t} - R_{f,t} = \alpha_j + b_j RMRF_t + s_j SMB_t + h_j HML_t + p_j PR1YR_t + e_{j,t}, \tag{7}$$

where $R_{j,t}$ is either a fund's net return (CRSP) or the holdings-based gross stock-portfolio return, and the factors are the market, size, value, and momentum factor-mimicking portfolios.

A **Jensen alpha** (single-factor CAPM) variant is also considered conceptually:

$$R_{j,t} - R_{f,t} = \alpha^J_j + \beta_j RMRF_t + u_{j,t}.$$

Interpretation. These regression intercepts adjust for covariation with priced risk factors, while the holdings-based CS measure adjusts returns by matching each stock to a characteristic benchmark.

4 How to read the empirical evidence (table-by-table interpretation)

4.1 Table I: Summary statistics for the merged database

Table I documents the construction of the merged database and describes the fund universe over time. The main facts are:

Panel A. Yearly Mutual Fund Universe Statistics

	Merged Database			Unmatched Funds		
Year	Number	TNA-Averaged Net Return (%/year)	Median TNA ($millions)	Number	TNA-Averaged Net Return (%/year)	Median TNA ($millions)
1975	241	30.9	35.5	0	—	—
1976	241	23.0	50.3	0	—	—
1977	226	−2.5	54.5	0	—	—
1978	222	9.0	49.0	0	—	—
1979	219	23.7	44.0	0	—	—
1980	364	31.3	48.9	0	—	—
1981	365	−2.7	44.8	0	—	—
1982	362	24.1	42.1	0	—	—
1983	347	20.4	52.9	0	—	—
1984	372	−0.1	80.3	0	—	—
1985	391	27.8	77.4	0	—	—
1986	418	15.8	98.2	0	—	—
1987	483	2.4	93.0	0	—	—
1988	543	15.9	83.8	0	—	—
1989	589	25.3	75.7	0	—	—
1990	637	−5.3	84.7	0	—	—
1991	679	32.8	78.5	11	25.2	22.4
1992	815	8.2	88.3	14	7.2	25.9
1993	949	14.2	100.1	31	15.6	9.5
1994	1,279	−1.6	98.5	54	−0.01	12.6
1975–1994	1,788	14.6	—	60	12.0	—

Panel B. Number of CRSP Mutual Fund Shareclasses in Merged Database

	Merged Database	
Year	Number of Funds	Number of Shareclasses
1975	241	241
1980	364	364
1985	391	391
1990	637	638
1991	679	684
1992	815	831
1993	949	996
1994	1,279	1,377

Panel C. Number of Funds and TNA-Average Stock holdings Percentage, by Investment Objective

	Universe		AG		G		G&I		I or B	
Year	Number	Stocks (%)	Number	Stocks (%)	Number	Stocks (%)	Number	Stocks (%)	Number	Stocks (%)
1975	241	79.9	67	86.9	76	89.1	52	83.8	46	51.0
1980	364	83.8	87	87.1	137	90.7	83	86.9	57	61.6
1985	391	85.4	85	93.4	151	89.5	102	82.0	53	60.8
1990	637	79.8	133	87.3	289	87.8	141	81.2	74	51.4
1994	1,279	82.7	201	92.7	703	90.7	246	82.5	129	54.5

Table I: Summary statistics for the merged database

			Merged Database				
Year	S&P 500 Return	CRSP VW Return	No.	TNA-Avg Gross Returns (%/year)	EW-Avg Gross Return (%/year)	TNA-Avg Net Return (%/year)	EW-Avg Net Return (%/year
1975	37.2	37.4	241	38.1	40.1	30.9	31.5
1976	23.8	26.8	241	26.7	28.0	23.0	23.6
1977	−7.2	−3.0	226	−3.0	0.2	−2.5	−0.1
1978	6.6	8.5	222	11.3	12.9	9.0	10.0
1979	18.4	24.4	219	27.9	32.9	23.7	26.2
1980	32.4	33.2	364	37.8	40.1	31.3	31.2
1981	−4.9	−4.0	365	−4.2	−2.3	−2.7	−0.6
1982	21.4	20.4	362	24.0	25.6	24.1	24.9
1983	22.5	22.7	347	23.6	23.9	20.4	20.1
1984	6.3	3.3	372	0.3	−0.6	−0.1	−0.8
1985	32.2	31.5	391	32.0	32.4	27.8	27.7
1986	18.5	15.6	418	17.7	15.8	15.8	14.1
1987	5.2	1.8	483	3.4	2.1	2.4	1.1
1988	16.8	17.6	543	18.7	18.2	15.9	14.5
1989	31.5	28.4	589	29.4	29.2	25.3	24.6
1990	−3.2	−6.0	637	−7.4	−7.4	−5.3	−5.5
1991	30.5	33.6	679	37.5	41.0	32.8	35.2
1992	7.7	9.0	815	9.1	10.0	8.2	9.1
1993	10.0	11.5	949	15.2	13.9	14.2	13.3
1994	1.3	−0.6	1,279	−0.4	−0.8	−1.6	−1.7
1975–1979	15.8	18.8	241	20.2	22.8	16.8	18.2
1980–1984	15.5	15.1	459	16.3	17.3	14.6	15.0
1985–1989	20.8	19.0	676	20.2	19.5	17.4	16.4
1990–1994	9.3	9.5	1,567	10.8	11.3	9.7	10.1
1975–1994	15.4	15.6	1,788	16.9	17.7	14.6	14.9

Table II: Mutual fund returns versus market indexes

- The merged sample grows dramatically in the number of funds from the 1970s into the 1990s.
- The table contrasts *matched* funds (with holdings + returns) and *unmatched* funds (in CRSP but not matched), which is important for understanding potential sample-selection issues.
- The table also reports how much of fund assets are held in stocks across investment objectives: even "equity" funds hold nontrivial nonstock positions (bonds/cash), which later matters for reconciling gross stock returns with overall net returns.

Practical implication: any comparison of holdings-based returns to fund net returns must recognize that funds are not 100% invested in equities.

4.2 Table II: Mutual fund returns versus market indexes

Table II provides the simplest performance comparison, by year and by subperiod:

- **Gross stock-portfolio returns** (computed from holdings) are higher than broad market indexes on average.
- **Net mutual fund returns** (what investors earn) are lower than market indexes on average.

Over the full sample, the TNA-weighted average gross stock-holdings return is about **16.9%** per year, while the S&P 500 and CRSP value-weighted indexes earn about **15.4–15.6%** per year. In contrast, the TNA-weighted average **net** mutual fund return is about **14.6%** per year.

Panel A. Performance of Mutual Fund Stock Portfolios (*continued*)

Year	Number	Merged Database TNA-Avg *CS* Measure (%/year)	EW-Avg *CS* Measure (%/year)
1989	589	−0.26	0.65
1990	637	−0.69	0.97
1991	679	1.95	1.74
1992	815	0.07	0.13
1993	949	1.70	1.02
1994	1,279	0.03	0.23
1975–1994†	1,788	0.71**	1.01***
		(2.79)	(3.76)

Panel B. Performance of Fund Stock Portfolios versus Net Fund Performance†

Period	Number	Merged Database TNA-Avg *CS* Measure (%/year)	EW-Avg *CS* Measure (%/year)	TNA-Avg $\alpha^{Net}_{Carhart}$ (%/year)	EW-Avg $\alpha^{Net}_{Carhart}$ (%/year)
1975–1979	241	1.17	1.62*	−0.81	−0.77
		(1.78)	(2.28)	(−1.04)	(−0.91)
1980–1984	459	0.71	1.00	−1.33	−1.14
		(1.18)	(1.51)	(−1.35)	(−1.12)
1985–1989	676	0.36	0.59	−0.73	−1.07**
		(1.15)	(1.35)	(−1.21)	(−2.01)
1990–1994	1,567	0.61	0.82**	−0.71	−0.47
		(1.19)	(2.77)	(−1.32)	(−0.98)
1975–1994	1,788	0.71**	1.01***	−1.16***	−1.15***
		(2.79)	(3.76)	(−2.96)	(−2.93)

Table III: Characteristic-adjusted performance and comparison with Carhart alphas

Economic interpretation. Managers appear to hold stocks that outperform, but investors do not capture this outperformance after costs and nonstock holdings are taken into account. This gap motivates the decomposition below.

4.3 Table III: Characteristic-adjusted performance and comparison with Carhart alphas

Panel A (CS measures). Panel A shows yearly characteristic-selectivity CS for the mutual fund stock portfolios. The average CS over 1975–1994 is positive:

- TNA-weighted CS is about **0.71% per year** (statistically significant),
- equal-weighted CS is about **1.01% per year** (statistically significant).

Interpretation. On average, funds select stocks that beat characteristic-matched benchmarks, suggesting stock-picking skill beyond static size/value/momentum exposures.

Panel B (CS vs. net Carhart alpha). Panel B contrasts the positive holdings-based CS measures with negative risk-adjusted *net* performance. The Carhart alpha estimated on net fund returns is around **-1.15% to -1.16% per year**, significantly below zero. **Interpretation.** Skill at picking stocks does not translate into positive net abnormal performance for investors once expenses, transaction costs, and nonstock holdings are included.

Panel A. Pearson Correlations

$\rho_{Pearson}$	CS	$\alpha^{Gross}_{Carhart}$	$\alpha^{Net}_{Carhart}$	α^{Gross}_{Jensen}	α^{Net}_{Jensen}
CS	1	—	—	—	—
$\alpha^{Gross}_{Carhart}$	0.57	1	—	—	—
$\alpha^{Net}_{Carhart}$	0.36	0.62	1	—	—
α^{Gross}_{Jensen}	0.58	0.74	0.43	1	—
α^{Net}_{Jensen}	0.33	0.49	0.84	0.63	1

Panel B. Spearman Rank Correlations

$\rho_{Spearman}$	CS	$\alpha^{Gross}_{Carhart}$	$\alpha^{Net}_{Carhart}$	α^{Gross}_{Jensen}	α^{Net}_{Jensen}
CS	1	—	—	—	—
$\alpha^{Gross}_{Carhart}$	0.65	1	—	—	—
$\alpha^{Net}_{Carhart}$	0.46	0.63	1	—	—
α^{Gross}_{Jensen}	0.53	0.67	0.44	1	—
α^{Net}_{Jensen}	0.40	0.49	0.80	0.68	1

Table IV: Correlations among performance measures

4.4 Table IV: Correlations among performance measures

Table IV reports cross-sectional correlations across funds between:

$$\text{CS}, \quad \alpha^{Gross}_{Carhart}, \quad \alpha^{Net}_{Carhart}, \quad \alpha^{Gross}_{Jensen}, \quad \alpha^{Net}_{Jensen}.$$

Key patterns:

- CS is *moderately* correlated with gross regression alphas (roughly 0.57–0.58). This means holdings-based characteristic adjustment and regression-based adjustment are related but not identical.
- Net performance measures (Carhart vs. Jensen) are *highly* correlated (e.g., Pearson $\approx$ 0.84), implying that most of the cross-fund variation in net alpha is captured similarly by one-factor and four-factor models.

Interpretation. The holdings-based CS provides a distinct lens on stock-picking: it conditions directly on stock characteristics rather than factor exposures inferred from returns.

4.5 Table V: Baseline return decomposition (industry + Vanguard comparison)

Table V is the core accounting exercise. It decomposes holdings-based gross stock returns into CS, CT, and AS, then places them alongside expenses and transaction costs to reconcile with net returns.

Panel A (mutual fund universe, TNA-weighted). The table reports:

$$\text{Gross stock return, CS, CT, AS, expense ratio, transaction costs, net return, turnover.}$$

The most important qualitative lessons:

Panel A. Mutual Fund Universe (TNA-Averaged)									
	Merged Database								
Year	Number	Gross Return (%/year)	CS (%/year)	CT^* (%/year)	AS^* (%/year)	Expense Ratio (%/year)	Transactions Costs (%/year)	Net Return (%/year)	Turnover (%/year)
1975	241	38.1	0.002	—	—	0.65	1.40	30.9	32.7
1976	241	26.7	0.23	1.15	25.2	0.64	1.32	23.0	32.6
1977	226	−3.0	0.34	0.52	−4.5	0.65	0.68	−2.5	29.3
1978	222	11.3	3.46	0.61	7.3	0.68	0.94	9.0	39.6
1979	219	27.9	1.79	−1.24	25.7	0.69	0.84	23.7	39.1
1980	364	37.8	0.83	0.13	35.2	0.70	0.96	31.3	53.6
1981	365	−4.2	0.21	−0.52	−4.1	0.70	0.91	−2.7	51.9
1982	362	24.0	2.53	1.83	19.1	0.74	1.23	24.1	60.7
1983	347	23.6	1.12	0.72	22.1	0.73	0.91	20.4	64.8
1984	372	0.3	−1.15	−0.63	2.6	0.80	1.02	−0.1	64.1
1985	391	32.0	−0.07	0.23	32.0	0.77	0.88	27.8	74.7
1986	418	17.7	0.40	−0.96	17.7	0.76	0.72	15.8	75.3
1987	483	3.4	1.54	0.78	1.8	0.81	0.56	2.4	79.2
1988	543	18.7	0.20	−1.07	18.9	0.88	0.56	15.9	68.1
1989	589	29.4	−0.26	0.31	29.7	0.88	0.50	25.3	65.1
1990	637	−7.4	−0.69	0.64	−7.0	0.91	0.47	−5.3	69.6
1991	679	37.5	1.95	−0.93	36.4	0.87	0.48	32.8	68.7
1992	815	9.1	0.07	−1.13	10.8	0.94	0.48	8.2	67.8
1993	949	15.2	1.70	0.22	12.1	0.96	0.49	14.2	71.7
1994	1,279	−0.4	0.03	−0.34	0.001	0.99	0.48	−1.6	72.8
1976–1979	241	15.7	1.46	0.26	13.4	0.67	1.04	13.3	35.2
1980–1984	459	16.3	0.71	0.31	15.0	0.73	1.01	14.6	59.0
1985–1989	676	20.2	0.36	−0.14	20.0	0.82	0.64	17.4	72.5
1990–1994	1,567	10.8	0.61	−0.31	10.5	0.93	0.48	9.7	70.2
1976–1994	1,788	15.8	0.75**	0.02	14.8	0.79	0.80	13.8	60.5

Panel B. Vanguard Index 500 Fund						
		Vanguard				
Year	S&P 500 Return (%/year)	Gross Return[T] (%/year)	Expense Ratio (%/year)	Transactions Costs[T] (%/year)	Net Return (%/year)	Turnover (%/year)
1975	37.2	—	—	—	—	—
1976	23.8	—	—	—	—	—
1977	−7.2	—	0.46	—	−7.8	—
1978	6.6	—	0.36	—	5.9	8
1979	18.4	—	0.30	—	18.1	29
1980	32.4	—	0.35	—	31.7	18
1981	−4.9	−4.5	0.42	0.07	−5.1	12
1982	21.4	22.1	0.39	0.12	21.0	11
1983	22.5	23.2	0.28	0.14	21.4	39
1984	6.3	7.1	0.27	0.09	6.3	14
1985	32.2	32.2	0.28	0.11	31.4	36
1986	18.5	18.5	0.28	0.05	18.1	29
1987	5.2	6.1	0.26	0.06	4.7	15
1988	16.8	16.6	0.22	0.05	16.2	10
1989	31.5	31.4	0.22	0.06	31.4	8
1990	−3.2	−3.6	0.22	0.05	−3.3	23
1991	30.5	30.3	0.20	0.05	30.2	5
1992	7.7	7.3	0.19	0.03	7.4	4
1993	10.0	10.0	0.19	0.02	9.9	6
1994	1.3	1.5	0.19	0.02	1.2	6
1977–1979	5.9	—	0.37	—	5.4	18.5†
1980–1984	15.5	12.0†	0.34	0.11†	15.1	18.8
1985–1989	20.8	21.0	0.25	0.07	20.4	19.6
1990–1994	9.3	9.1	0.20	0.03	9.1	8.8
1977–1994	13.7	14.2†	0.28	0.07†	13.3	16.1

Table V: Baseline return decomposition (industry + Vanguard comparison)

Fractile	Avg No	Gross Return (%/year)	*CS* (%/year)	*CT* (%/year)	*AS* (%/year)	Expense Ratio (%/year)	Trans. Costs (%/year)	Net Return (%/year)	$\alpha^{Net}_{Carhart}$ (%/year)	Turnover (%/year)
Top 10%	42	19.5	1.46*	0.28	16.8	0.97	2.65	15.5	−1.00	155
Top 20%	84	19.1	1.83***	0.31	16.3	0.94	2.07	15.9	−0.68	132
2nd 20%	84	17.3	1.33***	0.53*	15.1	0.90	1.12	15.2	−0.98*	82
3rd 20%	84	16.2	0.89**	0.16	15.1	0.82	0.82	14.3	−1.24**	57
4th 20%	84	15.8	0.59	−0.07	15.0	0.70	0.92	13.7	−1.40***	33
Bottom 20%	84	14.8	0.02	−0.06	14.6	0.64	0.33	13.2	−1.01**	18
Bottom 10%	42	15.2	0.24	0.05	14.6	0.69	0.28	13.4	−0.85*	14
Top-Bottom 10%	42	4.3*	1.22	0.23	2.2***	0.28***	2.37***	2.1	−0.15	141***
Top-Bottom 20%	84	4.3**	1.81***	0.37	1.7**	0.30***	1.74***	2.7**	0.33	114***
All Funds	420	16.2	0.79**	0.11	15.0	0.77	0.88	14.2	−1.12***	59

Table VI: Turnover-sorted decomposition

- **Selection** (CS) **is positive on average** and economically meaningful.
- **Timing** (CT) **is small on average**, suggesting limited ability to time characteristic payoffs.
- **Style** (AS) **accounts for the bulk of the gross stock return level** (as it must, since it reflects the typical equity premium plus characteristic premia).
- **Expenses + transaction costs are large enough to offset selection.**

A useful way to remember the economics is:

$$\underbrace{\text{positive stock selection}}_{\text{CS>0}} \quad \text{is roughly cancelled by} \quad \underbrace{\text{fees + trading frictions}}_{\text{expense ratio+trans. costs}}.$$

Nonstock holdings (why gross stock returns exceed net fund returns even after costs). The gap between gross stock-holdings returns and overall net fund returns is not *only* fees and trading costs. Funds also hold bonds and cash, whose average returns are typically lower than stock returns during this period. The paper's accounting attributes part of the gap to this *nonstock drag*.

Panel B (Vanguard Index 500 Fund). Panel B provides a benchmark: the Vanguard Index 500 fund has very low expense ratios and very low transaction costs (consistent with its low turnover). **Interpretation.** The comparison frames a key question: do active funds' higher gross returns and positive CS compensate for their higher costs relative to a passive index fund?

4.6 Table VI: Turnover-sorted decomposition (do frequent traders add value?)

Table VI sorts funds by prior-year turnover and reports average performance components in the subsequent year. The main monotonic patterns:

- **Higher turnover funds have higher gross stock-portfolio returns.**
- **Higher turnover funds have higher** CS **on average** (stronger stock-picking component).
- **Higher turnover funds have much higher transaction costs and somewhat higher expenses.**

The economic trade-off. High-turnover funds appear to generate higher gross performance partly through:

1. *style differences* (higher AS) and

2. *stronger selectivity* (higher CS),

but they also pay substantially higher trading costs. The table is therefore the clearest empirical expression of the paper's theme: *active management can create value at the stock-selection level, but that value is partially (or fully) absorbed by implementation costs.*

Risk-adjusted net performance. Even when high-turnover funds have higher *net* raw returns than low-turnover funds, the net Carhart alphas reported are still generally negative across the turnover spectrum. This is consistent with the idea that part of the raw-return advantage reflects exposure to compensated characteristics/factors rather than pure abnormal performance.

5 Integrated intuition and what to take away for further work

5.1 A "mechanism" summary

This paper's decomposition implies a clean economic mechanism:

1. **Managers exhibit stock-selection skill** in the sense of choosing stocks that outperform characteristic-matched benchmarks (CS > 0).

2. **Characteristic timing is weak** on average (CT ≈ 0).

3. **Implementation is costly**: expenses and transaction costs are large enough to offset the selection advantage.

4. **Portfolio composition matters**: nonstock holdings further reduce realized fund returns relative to an all-equity gross stock portfolio.

5.2 Why this paper is methodologically important

The main methodological contribution is the *holdings-based performance decomposition*:

- It links performance to *what managers hold* rather than only to return covariances.

- It clarifies whether underperformance arises from lack of selection skill or from high costs.

- It creates objects that can be mapped to IO-style ideas about *production of returns* and *cost of implementation*, which is useful for connecting asset pricing to institutional features (trading technology, fees, liquidity).

5.3 Caveats

- The CS benchmark controls for size, book-to-market, and momentum, but other characteristics (e.g. liquidity) may matter; thus, measured selectivity may not fully isolate pure stock-picking.
- Transaction costs are estimated from a model calibrated on institutional trades; there is estimation error and potential mismatch to mutual fund execution.
- Holdings are observed quarterly, so intra-quarter trading and timing are not fully captured.